# Lathiesh Mahendran

AI & Full-Stack Engineer

Portfolio lathiesh-portfolio.vercel.app

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## ABOUT

AI engineer with 3+ years at early-stage startups, building LLM-powered products from idea to production. I work across the whole system: architecture, model selection, retrieval, agents and tool calling, evaluation, and cost and latency. I do my best work when I can own a problem end to end.

## SKILLS

**AI & LLM:** Anthropic Claude API, OpenAI API, Google Vertex AI (Gemini), AWS Bedrock, AI agents and tool calling, MCP, Vercel AI SDK, LangChain, Exa AI, Parallel AI, Olostep, Riveter, LLM evaluation
**RAG & Retrieval:** Pinecone, hybrid search with Reciprocal Rank Fusion, LLM-based metadata filtering, ThirdAI NeuralDB
**Backend & Data:** Python, FastAPI, Django, SQLAlchemy, Neon (PostgreSQL), MongoDB, PropelAuth, REST APIs
**Data Providers & APIs:** Crustdata, HubSpot CRM, Google Maps / Places / Geocoding, SEC EDGAR, GLEIF, EPA
**Infra & Tools:** AWS Lambda, Trigger.dev, Render, Azure, Doppler, GitHub Actions, Grafana, Pytest, Vitest
**Frontend:** React, TypeScript, Next.js, Tailwind CSS, TanStack Query, Plate.js
**AI-Assisted Engineering:** Claude Code, Codex

## EXPERIENCE

**Product Engineer (AI & Full-Stack)** Corvus AI (New York, USA · Remote) **11/2025 – Present**

*5-person AI startup building sales intelligence for the chemical industry; I work directly with the CEO and CTO.*

- **Architected the product's multi-step agentic research pipeline:** tool-using LLM agents (**Vertex AI Gemini**, **OpenAI**, **Claude**, **Exa AI**) on **Trigger.dev** take a company name and produce a cited report on its products, facilities and operations. Stages run in parallel and pick up where they left off after a failure.
- **Engineered enrichment and classification pipelines over a ~200K-company dataset** with **Crustdata**, **Exa AI**, **Olostep**, **Riveter**, **OpenAI** and **Gemini**, covering batch and real-time processing, entity matching and deduplication.
- **Built LLM evaluation harnesses** (ground-truth datasets, precision/recall scoring) so model and prompt changes are benchmarked before they reach production. Brought LLM cost down with prompt caching, right-sizing models per stage, and cost telemetry.
- **Extended the agent harness** behind the product's **Claude**-based AI agent (**Vercel AI SDK**, tool calling), working on how it manages context and persists state, and built **MCP** tools that give research agents access to EPA facility data.
- **Shipped full-stack features** (**FastAPI**, **Neon PostgreSQL**, **React/TypeScript**, **Render**): **PropelAuth** multi-tenant auth, **Google Maps/Places** geospatial prospecting, a **Plate.js** live editor, PDF/CSV/email document pipelines, **Parallel AI** news monitoring and admin tooling.
- **Brought production error alerts to zero** by setting up **Slack** and **Grafana** monitoring across services and pipelines, then fixing what it surfaced through backfills, data repairs and job recovery.
- **Automated engineering and data operations with Claude Code and Codex:** custom agent skills, operational runbooks, and parallel sub-agent workflows that audited thousands of production records.

**Software Developer** The Yellow Network **08/2023 – 11/2025**

*Early-stage startup building Nifo, an AI co-innovation workflow platform; worked directly with the founder.*

- **Co-built Tia from scratch with one other engineer:** the RAG assistant live on tvsscs.com for **TVS Supply Chain Solutions** (NSE-listed). It answers visitor support questions and captures sales-qualified leads into **HubSpot CRM** for the sales team.
- **Designed Tia's serverless architecture** on **AWS Lambda**, **Pinecone** and **AWS Bedrock** with tuned cold starts. Hybrid retrieval merges semantic and targeted service queries via **Reciprocal Rank Fusion**; a fast model extracts Pinecone metadata filters and a stronger model writes the answer, keeping cost and latency down.
- **Built a 150-query retrieval evaluation suite** for Tia tracking success rate, response latency, link relevance and category coverage, and used the failing queries to guide each round of retrieval fixes.
- **Co-built Nifo end to end** as one of two core developers: a **Django** and **PostgreSQL** backend and a **Next.js/TypeScript** frontend, with secure authentication and per-user access control.
- **Developed RAG-based AI agents** for document generation, enterprise insights, and solution and provider search, using **LangChain**, **OpenAI** and open-source LLMs, with **ThirdAI NeuralDB** for semantic indexing and retrieval.
- **Deployed and scaled LLM endpoints** on **Azure AI Foundry** and **Azure VMs**, and built **Selenium** scraping and preprocessing pipelines that produced structured, LLM-ready datasets.

## PROJECTS

**Saramsa.ai** AI feedback-to-roadmap platform · saramsa.ai

Built and launched an LLM platform that pulls customer feedback from **Slack**, email, **Jira**, **Azure DevOps**, **Asana** and **Linear**, groups it into themes and turns it into prioritized roadmap tasks (**OpenAI**, **Next.js**, **FastAPI**, **Neon**, **Azure**).